The documentation process conversion measures necessary to comply with this revision shall be completed by 25 February 1998

INCH-POUND

MIL-PRF-19500/408E
25 November 1997
SUPERSEDING
MIL-S-19500/408D
25 August 1992

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, HIGH-POWER
TYPES 2N3715 AND 2N3716 JAN, JANTX, JANTXV AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, high-power transistors. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 similar to TO-3.

1.3 Maximum ratings.

	P_T 1/ T_A = +25°C	P_T T_C = +100°C	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_J and T_{STG}	$R_{\theta JC}$
	W	W	V dc	V dc	V dc	A dc	A dc	°C	°C/W max
2N3715	5.0	85.7	80	60	7.0	4.0	10	-65 to +200	1.17
2N3716	5.0	85.7	100	80	7.0	4.0	10	-65 to +200	1.17

1/ Derate linearly 28.57 mW/°C above T_A = +25°C.

2/ Derate linearly 0.857 W/°C above T_C = +100°C.

1.4 Primary electrical characteristics.

	h_{FE1} 1/	h_{FE4} 1/	$V_{BE(SAT)1}$ 1/	$V_{CE(SAT)1}$ 1/	C_{obo}	$\|h_{fe}\|$
	V_{CE} = 2.0 V dc I_C = 3.0 A dc	V_{CE} = 4.0 V dc I_C = 10 A dc	I_C = 5.0 A dc I_B = 0.5 A dc	I_C = 5.0 A dc I_B = 0.5 A dc	V_{CB} = 10 V dc I_E = 0 f = 1 MHz	V_{CE} = 10 V dc I_C = 0.5 A dc f = 1 MHz
			V dc	V dc	pF	
Min	30	5.0				4.0
Max	120		1.5	1.0	500	20

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 herein

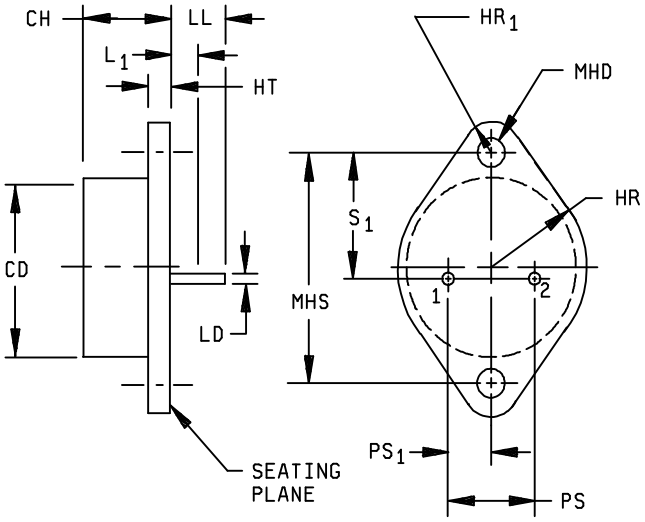
3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750, MIL-PRF-19500, and herein.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I herein.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3 herein.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	- - -	.875	- - -	22.22	
CH	.270	.380	6.86	9.65	
HT	.060	.055	1.52	1.40	
HR	.495	.525	12.57	13.34	
HR_1	.131	.188	3.33	4.78	
LD	.038	.043	0.97	1.09	7
LL	.312	.500	7.92	12.70	
L_1	- - -	.050	- - -	1.27	
MHD	.151	.165	3.84	4.19	
MHS	1.177	1.197	29.90	30.40	
PS	.420	.440	10.67	11.18	4
PS_1	.205	.225	.521	5.72	4
s_1	.655	.675	16.64	17.15	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Terminal 1, emitter; terminal 2,base; case, collector.
4. These dimensions should be measured at points .050-.055 inch (1.27-1.40 mm) below seating plane. When gauge is not used, measurement will be made at the seating plane.
5. The seating plane of the header shall be flat within .004 inch (0.10 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .006 inch (0.15 mm) concave to .006 inch (0.15 mm) convex overall.
6. Collector shall be electrically connected to the case.
7. LD applies between L_1 and LL. Diameter is uncontrolled in L_1.

FIGURE 1. Physical dimensions of transistor types 2N3715 and 2N3716 (TO-3).

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.3 Screening (JANS, JANTX and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Thermal response (see 4.3.2)	Thermal response (see 4.3.2)
9	I_{CES1} and h_{FE2}	I_{CES1}
11	I_{CES1} and h_{FE2}; ΔI_{CES1} = 100 percent of initial value or 1 μA dc, whichever is greater. Δh_{FE2} = ± 15 percent of initial value.	I_{CES1} and h_{FE2}; ΔI_{CES1} = 100 percent of initial value or 1 μA dc, whichever is greater.
12	See 4.3.1	See 4.3.1
13	ΔI_{CES1} = 100 percent of initial value or 1 μA dc, whichever is greater; Δh_{FE2} = ± 15 percent of initial value; subgroup 2 of table I herein	ΔI_{CES1} = 100 percent of initial value or 1 μA dc, whichever is greater; Δh_{FE2} = ± 20 percent of initial value; subgroup 2 of table I herein

1/ This test shall be performed any time before screen 3.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_J = 187.5 ± 12.5°C; V_{CE} = 25 ± 5 V dc; T_A = ≤ 100°C.

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4.3.2 Thermal response (ΔV_{BE}) measurements. The ΔV_{BE} measurements shall be performed in accordance with method 3131 of MIL-STD-750. The ΔV_{BE} conditions (I_H and V_H) and maximum limit shall be derived by each vendor. The chosen ΔV_{BE} measurement and conditions for each device in the qualification lot shall be submitted in the qualification report and a thermal response curve shall be plotted. The chosen V_{BE} shall be plotted. The chosen V_{BE} shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. One hundred percent safe operating area (SOA) testing may be performed in lieu of thermal response testing herein provided that the appropriate conditions of temperature, time current and voltage to achieve die attach integrity are submitted to the qualifying activity. The following parameter measurements shall apply:

a. I_M ..... 10 mA.

b. V_{CE} measurement voltage ..... 20 V (same as V_H).

c. I_H collector heating current ..... 4 A (minimum).

d. V_H collector-emitter heating voltage ..... 20 V (minimum).

e. t_H heating time ..... 100 ms.

f. t_{MD} measurement delay time ..... 50 μs to 80 μs.

g. t_{SW} sample window time ..... 10 μs (maximum).

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection . Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIa (JANS) and table Vib (JANTX and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, appendix E, table VIa of MIL-PRF-19500.

Subgroup	Method	Conditions
B4	1037	V_{CB} = 30 V dc, P_T = 4 W at T_A = +25°C ± 3°C, t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced air on the device shall be permitted.
B5		A separate sample may be pulled for each test.
	1027	V_{CB} = 30 V dc, T_A = +125°C ± 25°C for 96 hours; P_T = 4 W at T_A = +100°C or adjusted as required by the chosen T_A to give an average lot T_J = 225°C. Marking legibility requirement shall not apply.
B6	3131	See 4.5.2.

4.4.2.1 Group B inspection, appendix E, table VIb of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	1037	V_{CB} ≥ 10 V dc; ΔT_J between cycles ≥ 100°C; t_{on} = t_{off} ≥ 1 minute for 2,000 cycles. No heat sink or forced air cooling on the device shall be permitted.
B5	3131	See 4.5.2.
B6	1032	T_A = +200°C.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Conditions
C2	1056	Test condition B.
C2	2036	Test condition A, weight = 10 lbs, application time = 15 seconds.
C6	1037	$V_{CB} \geq 10$ V dc; ΔT_J between cycles $\geq 100°C$, $t_{on} = t_{off} \geq 1$ minute for 6,000 cycles. No heat sink or forced air cooling on the device shall be permitted.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with test method 3131 of MIL-STD-750. The following details shall apply.

a. Collector current magnitude during power application shall be 1.0 A dc.

b. Collector to emitter voltage magnitude shall be 10 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference temperature measuring point shall be within the range $25°C \leq T_R \leq 75°C$ and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to header.

f. Maximum limit of $R_{\theta JC}$ shall be 1.17°C/W.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Collector-emitter breakdown voltage	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N3715				60		V dc
2N3716				80		V dc
Collector-base cutoff current	3036	Base condition D	I_{CBO}			
		V_{CB} = 80 V dc			10	μA dc
2N3715					10	μA dc
2N3716		V_{CB} = 100 V dc				
Emitter-base cutoff current	3061	Base condition D; V_{EB} = 7 V dc	I_{EBO}		1.0	mA dc
Collector-emitter cutoff current	3041	Bias condition A	I_{CEX1}			
		V_{BE} = 1.5 V dc; V_{CE} = 60 V dc;				
2N3715					1.0	ma dc
2N3716		V_{BE} = 1.5 V dc; V_{CE} = 80 V dc			1.0	mA dc
Collector-emitter cutoff current	3041	Bias condition C	I_{CES1}			
		V_{CE} = 60 V dc			1.0	mA dc
2N3715					1.0	mA dc
2N3716		V_{CE} = 80 V dc				
Base-emitter saturated voltage	3066	Test condition A; I_C = 5 A dc; I_B = 0.5 A dc; pulsed (see 4.5.1)	$V_{BE(SAT)1}$		1.5	V dc
Base-emitter saturated voltage	3066	Test condition A; I_C = 10 A dc; I_B = 2 A dc; pulsed (see 4.5.1)	$V_{BE(SAT)2}$		3.0	V dc
Collector-emitter saturated voltage	3071	I_C = 5 A dc; I_B = 0.5 A dc; pulsed (see 4.5.1)	$V_{CE(SAT)1}$		1.0	V dc
Collector-emitter saturated voltage	3071	I_C = 10 A dc; I_B = 2 A dc; pulsed (see 4.5.1)	$V_{CE(SAT)2}$		2.5	V dc
Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 1.0 A dc; pulsed (see 4.5.1)	h_{FE1}	50	150	
Forward current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 3.0 A dc; pulsed (see 4.5.1)	h_{FE2}	30	120	

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 5 A dc; pulsed (see 4.5.1)	h_{FE3}	10		
Forward-current transfer ratio	3076	V_{CE} = 4.0 V dc; I_C = 10 A dc; pulsed (see 4.5.1)	h_{FE4}	5		
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition C	I_{CES2}			
2N3715		V_{CE} = 50 V dc			5.0	mA dc
2N3716		V_{CE} = 70 V dc			5.0	mA dc
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 3.0 A dc; pulsed (see 4.5.1)	h_{FE5}	12		
Subgroup 4						
Switching parameters						
Pulse delay time		See figure 2	t_d		0.2	µs
Pulse rise time		See figure 2	t_r		1.3	µs
Pulse storage time		See figure 2	t_s		1.2	µs
Pulse fall time		See figure 2	t_f		1.2	µs
t_{off}		$t_s + t_f$	t_{off}		2.0	µs
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 0.5 A dc; f = 1 kHz	h_{fe}	30	300	
Magnitude of small-signal short-circuit forward current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 0.5 A dc; f = 100 kHz - 1 MHz	$\|h_{fe}\|$	4.0	20	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; f = 1 MHz	C_{obo}		500	pF

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +25°C; t ≥ 1 s; 1 cycle; (see figure 3)				
Test 1		V_{CE} = 15 V dc; I_C = 10 A dc				
Test 2		V_{CE} = 40 V dc; I_C = 3.75 A dc				
Test 3						
2N3715 2N3716		V_{CE} = 55 V dc; I_C = 0.9 A dc V_{CE} = 65 V dc; I_C = 0.9 A dc				
Safe operating area (clamped inductive)		T_A = +25°C; I_C = 10 A dc; V_{CC} = 15 V dc; (see figures 5 and 6)				
2N3715 2N3716		Clamp voltage = 60 V dc Clamp voltage = 80 V dc				
Test 1		t_p approximately 1.0 ms (vary to obtain I_C); R_{BB1} = 5Ω; V_{BB1} = 8 V dc; R_{BB2} = 4; V_{BB2} = 0; V_{CC} = 15 V dc; I_C = 5 A dc; L = 4 mH; Q ≥ 100 at 1 kHz; .05 ohms, 20 A dc; (Stanford Miller CK-20 or equivalent)				
Test 2		t_p approximately 2.2 ms (vary to obtain I_C); R_{BB1} = 5Ω; V_{BB1} = 8 V dc; R_{BB2} = 4; V_{BB2} = 0; V_{CC} = 15 V dc; I_C = 2.2 A dc; L = 20 mH; Q ≥ 6000 at 10 kHz; 0.22 ohms, 12.5 A dc; (Two Stancor C-2688 in series or equivalent)				
Electrical measurements		See table II, steps 1 and 4.				
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Groups A, B, and C electrical end-point inspection measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector-emitter cutoff current	3041	Bias condition C	I_{CES1}			
	2N3715		V_{CE} = 50 V dc			10	µA dc
	2N3716		V_{CE} = 70 V dc			10	µA dc
2.	Collector-emitter (voltage saturated)	3071	I_C = 5 A dc; I_B = 0.5 A dc; pulsed (see 4.5.1)	$V_{CE(SAT)1}$		1.0	V dc
3.	Base-emitter voltage saturated	3066	Bias condition A; I_C = 5.0 A dc; I_B = 0.5 A dc; pulsed (see 4.5.1)	$V_{BE(SAT)1}$		1.5	V dc
4.	Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 3.0 A dc; pulsed (see 4.5.1)	h_{FE2}	30	120	
5.	Collector-emitter cutoff current	3041	Bias condition C	ΔI_{CES1} 4/	100 percent of initial value or 1 µA dc; whichever is greater.		
	2N3715		V_{CE} = 50 V dc				
	2N3716		V_{CE} = 70 V dc				
6.	Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 3.0 A dc; pulsed (see 4.5.1)	Δh_{FE2} 4/	± 25 percent change from initial value		
7.	Collector-emitter (voltage saturated)	3071	I_C = 5 A dc; I_B = 0.5 A dc; pulsed (see 4.5.1)	$\Delta V_{CE(SAT)1}$ 4/	± 50 percent mV dc change from previously measured value.		
8.	Thermal response	3131	See 4.3.2	ΔV_{BE}			

See footnotes at end of table.

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TABLE II. Groups A, B, and C electrical end-point inspection measurements 1/ 2/ 3/ - Continued.

1/ The electrical measurements for appendix E, table VIa (JANS) of MIL-PRF-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1, 2, 3, and 4.

b. Subgroup 4, see table II herein, steps 1, 2, 3, 4, 7, and 8.

c. Subgroup 5, see table II, steps 1, 2, 3, 4, 5, 6, and 7.

2/ The electrical measurements for appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1, and 3.

b. Subgroup 3, see table II herein, steps 1, 3, 4, 5, and 6.

c. Subgroup 6, see table II herein, steps 1, 4, 5, and 6.

3/ The electrical measurements for appendix E, table VII of MIL-PRF-19500 are as follows:

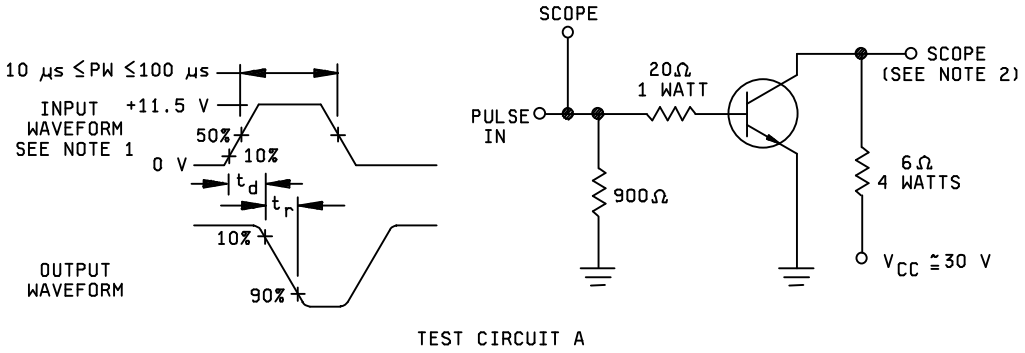
a. Subgroup 2, see table II herein, steps 1, 2, 3, and 4 (JANS); and 1, 2, and 4 (JANTX and JANTXV).

b. Subgroup 3, see table II herein, steps 1, 2, 3, and 4 (JANS); and 1, 2, and 4 (JANTX and JANTXV).

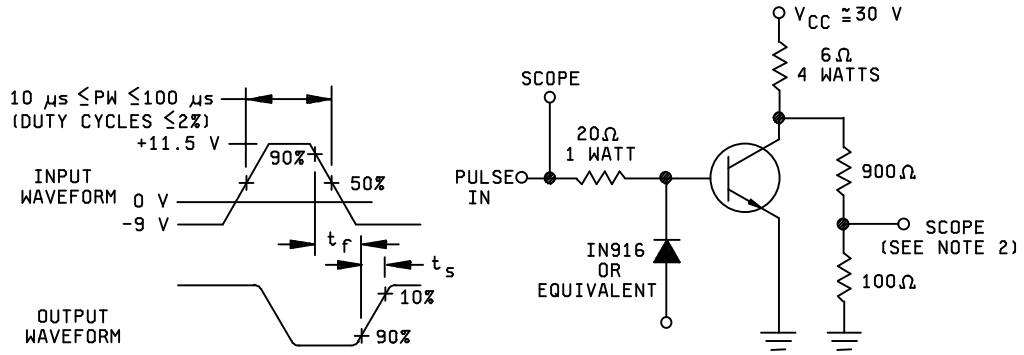
c. Subgroup 6, see table II herein, steps 1, 2, 3, 4, 5, 6, 7, and 8 (JANS); and 1, 2, 3, 4, 5, and 6 (JANTX and JANTXV).

4/ Devices which exceed the group A limits for this test shall not be acceptable.

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TEST CIRCUIT A



TEST CIRCUIT B

NOTES:

1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 20.0$ ns, $t_f \leq 1$ µs, 10 µs ≤ PW ≤ 100 µs, $Z_{OUT} = 50\Omega$, duty cycle ≤ 2 percent.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 5$ ns, $Z_{IN} \geq 100$ kΩ, $C_{IN} \leq 12$ pF.
3. Test circuit A for t_d and t_r; test circuit B for t_s and t_f.

FIGURE 2. Pulse response test circuits.

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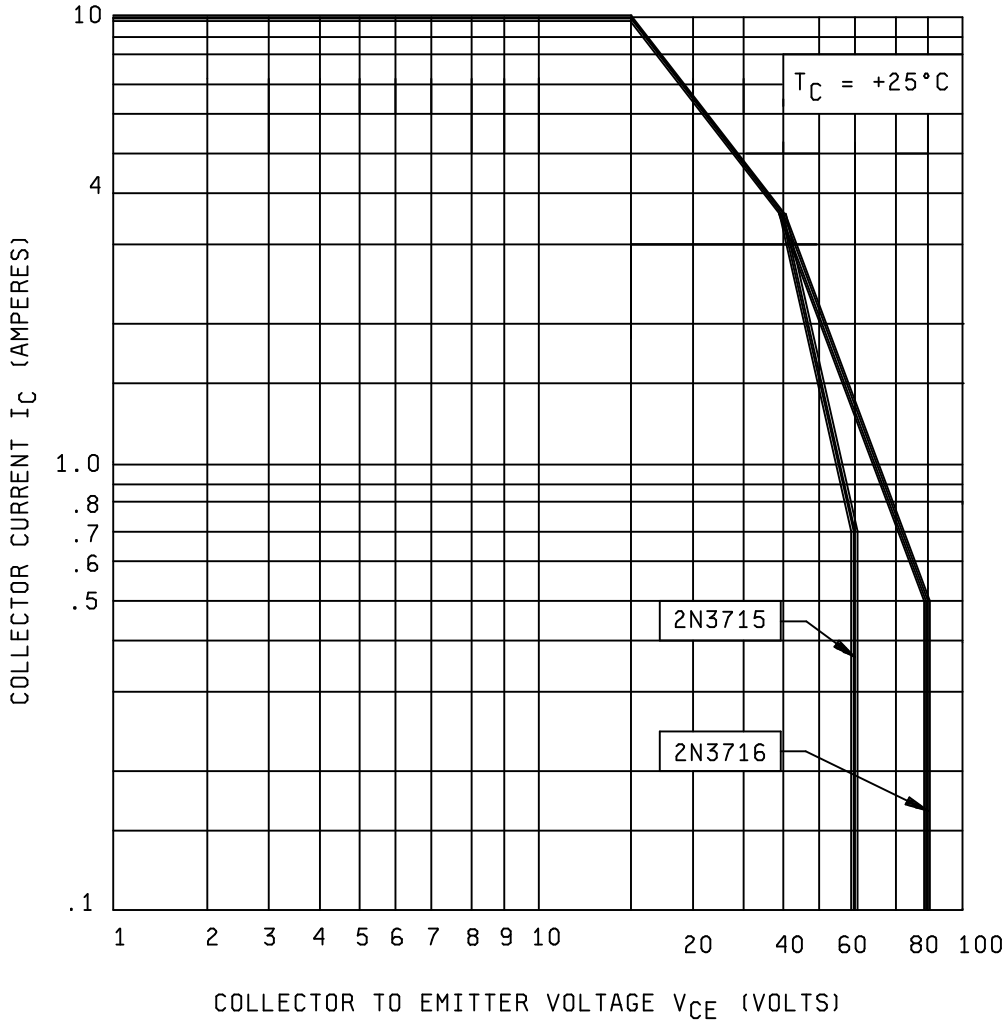


FIGURE 3. Maximum safe operating graph (dc).

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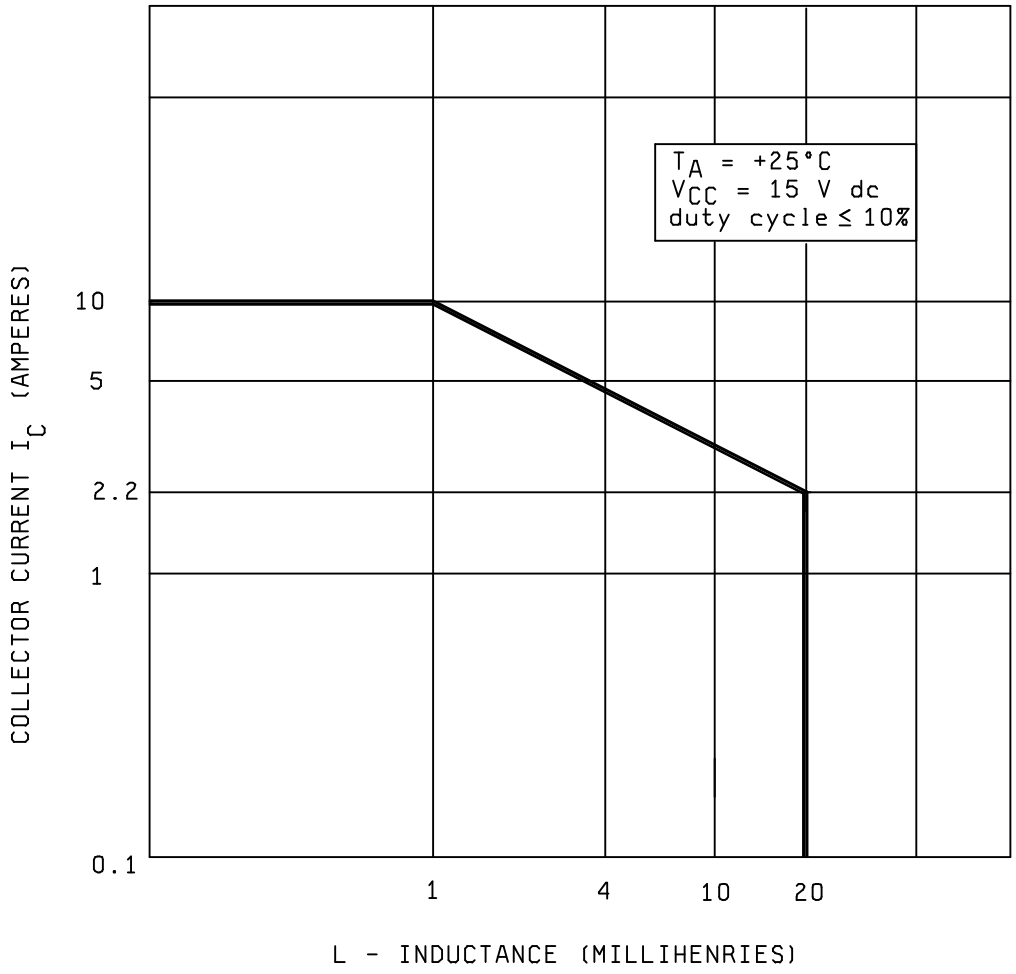


FIGURE 4. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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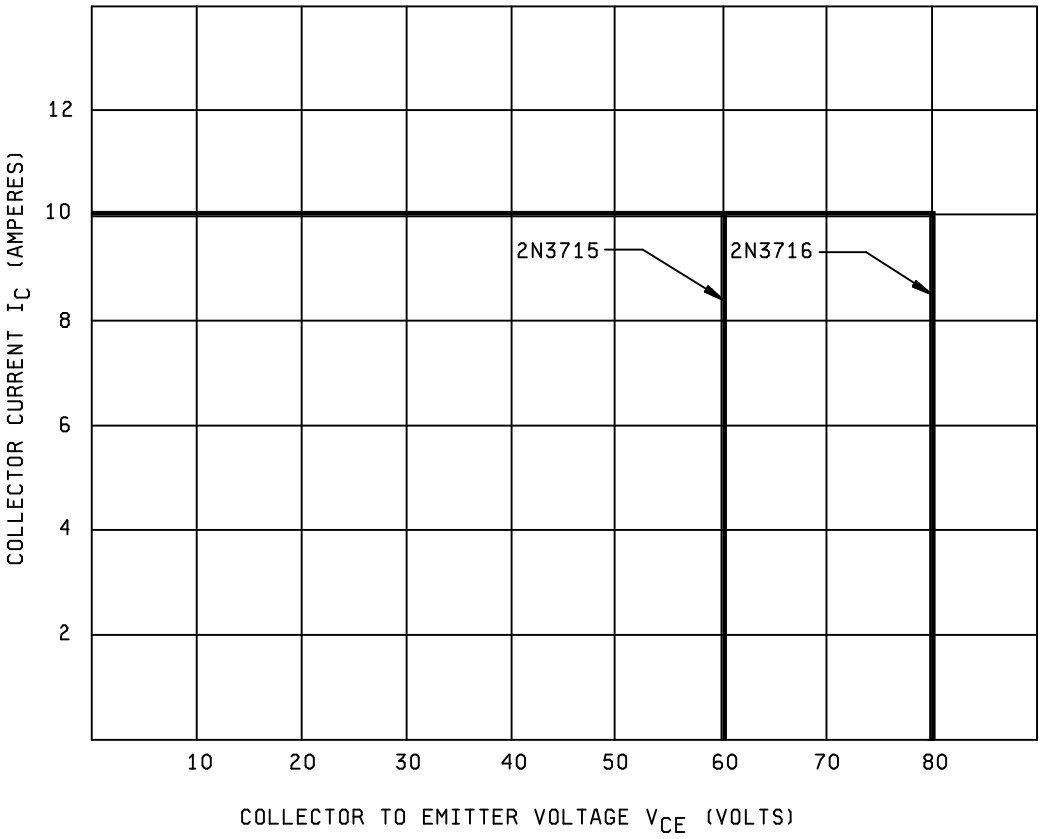
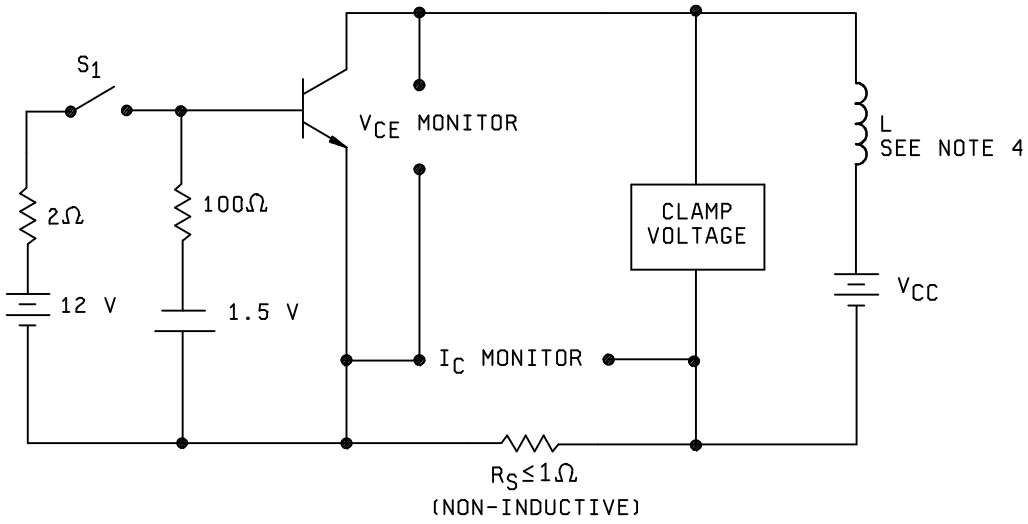


FIGURE 5. Safe operating area for switching between saturation and cutoff (clamped inductive load).

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Procedure:

1. With switch S1 closed, set the specified test conditions.
2. Open S1. Device fails if clamp voltage not reached.
3. Perform specified end-point tests.
4. L = 4.0 mH, 0.05Ω, 20 A, Q ≥ 100 at 1 kHz, (Sanford Miller CK-20 or equivalent)

FIGURE 6. Clamp inductive sweep test circuit.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Interchangeability information. MIL-PRF-19500/621 is a T0-254 package version of MIL-PRF-19500/379, which is a T0-3 package version. The military 2N7369 contains the same die as the military 2N3792. The MIL-PRF-19500/621 is preferred over the MIL-PRF-19500/379 whenever interchangeability is not a problem. For new design use 2N7369. The 2N2792 is inactive for new design.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, AV, MI
Navy - AS, CG, MC
Air Force - 13, 19, 85, 99

Preparing activity:
DLA - CC

(Project 5961- 1902-13)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/408E	2. DOCUMENT DATE (YYMMDD) 971125

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICES, TRANSISTOR, NPN, SILICON, HIGH-POWER, TYPES 2N3715 AND 2N3716, JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial (614)692-0510 (2) AUTOVON 850-0510
c. ADDRESS (Include Zip Code) Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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